4.2.1 Sleep Quality Classification Standards

Scientific Foundation

This system's classification is based on established sleep research and clinical guidelines:

Key References:

- Okamoto-Mizuno & Mizuno (2012) - Thermal environment effects on sleep [1]
- WHO Guidelines for Community Noise (1999) - Sleep disturbance thresholds [2]
- National Sleep Foundation (2023) - Bedroom environment standards [3]
- Ancoli-Israel et al. (2003) - Actigraphy-based sleep assessment [4]
- Buysse et al. (1989) - Pittsburgh Sleep Quality Index (PSQI) [5]

Parameter Standards

Temperature

Optimal: 15-19°C (60-67°F)

Rationale: Body temperature drops 1-2°C during sleep for optimal rest [1]. Temperatures outside this range disrupt thermoregulation and reduce REM/deep sleep.

Thresholds:

- 15-19°C: No deduction
- 12-24°C: -5 points
- <12°C or >24°C: -10 points
- Variance >2°C during night: -15 points

Sound Level

Optimal: <30 dB (quiet bedroom)

Rationale: WHO guidelines show noise >30 dB causes sleep fragmentation and autonomic arousal [2].

Thresholds:

- <30 dB average: No deduction
- 30-45 dB: -5 points
- >45 dB: -10 points
- Peak >70 dB (snoring): -5 points each (max -20)

Motion/Restlessness

Normal: <5 movements/hour

Rationale: Actigraphy studies show excessive motion indicates inability to reach deep sleep [4].

Thresholds:

- <5 events/hour: No deduction
- Each motion event: -2 points
- >10 events total: Additional -10 penalty

Humidity

Optimal: 30-50% relative humidity

Rationale: Affects respiratory comfort and mucous membrane function [3].

Thresholds:

- 30-50%: No deduction
- 25-60%: -5 points
- <25% or >60%: -10 points

Scoring Algorithm

Base Score: 100 points
Final Score = Base - (Temperature + Sound + Motion + Humidity deductions)
Score clamped to [0, 100]

Classification:
• Score ≥ 60: "Good Sleep" ■
• Score < 60: "Poor Sleep" ■■

60-point threshold rationale: Aligns with PSQI clinical standard (score <5 = good sleep) [5]. Represents meeting 60% of optimal environmental conditions.

Detailed Scientific Basis

Temperature Monitoring - Extended Rationale

The human body undergoes natural thermoregulation during sleep, with core body temperature dropping by approximately 1-2°C to facilitate sleep onset and maintenance (Okamoto-Mizuno &

Mizuno, 2012). Ambient temperature significantly affects this process.

Research Findings:

- Temperatures above 24°C (75°F) increase wakefulness and decrease REM and slow-wave sleep
- Temperatures below 12°C (54°F) cause sleep disruption and frequent arousals
- Optimal thermal comfort zone for sleep: 16-19°C with individual variation of ±2°C

Sound Level Monitoring - Extended Rationale

Environmental noise is a significant sleep disruptor, causing autonomic nervous system activation, increased cortisol levels, and sleep fragmentation even without conscious awakening (Halperin, 2014).

WHO Guidelines (1999) - Sleep Disturbance Thresholds:

- <30 dB: No substantial biological effects
- 30-40 dB: Sleep disturbance possible in sensitive individuals
- 40-55 dB: Moderate sleep disturbance, increased body movements
- >55 dB: High probability of awakening and sleep fragmentation

Peak Noise Events: Sound peaks >70 dB (e.g., snoring, door slams) cause micro-arousals and prevent deep sleep stage progression, even if the individual doesn't fully wake.

Motion Detection - Extended Rationale

Actigraphy (motion-based sleep assessment) is a validated method for evaluating sleep quality, with strong correlation to polysomnography (Ancoli-Israel et al., 2003). Excessive movement indicates:

- Difficulty achieving or maintaining deep sleep stages
- Sleep fragmentation and frequent stage transitions
- Potential sleep disorders (e.g., periodic limb movement disorder, restless legs syndrome)

Research Standards:

- Normal sleepers: 10-20 movements per 8-hour sleep period (1.25-2.5/hour)
- Poor sleepers: >40 movements per 8-hour period (>5/hour)
- Each movement represents potential brief arousal or sleep stage disruption

Humidity Monitoring - Extended Rationale

Indoor humidity significantly affects respiratory comfort, mucous membrane hydration, and subjective sleep quality (Pan et al., 2018; National Sleep Foundation, 2023).

Health Impacts:

- Low Humidity (<30%): Dry throat, nasal congestion, respiratory irritation, increased susceptibility to infections

• High Humidity (>60%): Difficulty breathing, mold growth risk, perceived stuffiness, thermal discomfort

• Optimal (30-50%): Comfortable breathing, proper mucous membrane function

Implementation Approach

Phase 1: Rule-Based (Current)

Deterministic scoring using research thresholds. No training data required. Transparent and clinically interpretable. Suitable for Basic-level requirements.

Phase 2: ML Enhancement (Future)

- Collect patient-reported sleep quality (PSQI surveys)
- Train Random Forest model on labeled data
- Target: >75% accuracy vs. patient ratings
- Validate against clinical assessments

Limitations

- **Environmental only:** Does not capture physiological signals (heart rate, brain activity)
- **Not diagnostic:** Screening tool, not a replacement for polysomnography
- **Population averages:** Individual optimal conditions may vary
- **Missing context:** Doesn't account for stress, medications, health conditions

Clinical Use: Results inform environmental optimization, not medical diagnosis.

Composite Sleep Quality Scoring Formula

Detailed Algorithm:

```
Base Score: 100 points

Temperature Component:
IF temp in [15-19°C]: 0 deduction
ELSE IF temp in [12-24°C]: -5 points
ELSE: -10 points

IF temp_variance > 2°C: -15 points

Sound Component:
IF avg_sound < 30 dB: 0 deduction
ELSE IF avg_sound < 45 dB: -5 points
ELSE: -10 points

FOR each sound_peak > 70 dB:
deduct -5 points (max total: -20 points)

Motion Component:
FOR each motion_event:
```

```
deduct -2 points

IF total_motion_events > 10:
additional -10 points penalty

Humidity Component:
IF humidity in [30-50%]: 0 deduction
ELSE IF humidity in [25-60%]: -5 points
ELSE: -10 points

Final Score = Base Score - Total Deductions
Final Score = CLAMP(Final Score, 0, 100)
```

Quick Reference Table

Parameter	Optimal	Acceptable	Poor	Score Impact
Temperature	15-19°C	12-24°C	<12 or >24°C	0 / -5 / -10
Temp Variance	<2°C	-	>2°C	0 / -15
Sound (avg)	<30 dB	30-45 dB	>45 dB	0 / -5 / -10
Sound (peak)	<70 dB	-	>70 dB	-5 each (max -20)
Motion	<5/hr	-	>10 total	-2 each + penalty
Humidity	30-50%	25-60%	<25 or >60%	0 / -5 / -10

Classification: Good Sleep: ≥60 points | Poor Sleep: <60 points

References (Full Citations)

[1] Okamoto-Mizuno, K., & Mizuno, K. (2012). Effects of thermal environment on sleep and circadian rhythm. *Journal of Physiological Anthropology*, 31(1), 14. https://doi.org/10.1186/1880-6805-31-14

[2] World Health Organization. (1999). *Guidelines for Community Noise*. Chapter 4: Sleep Disturbance, pp. 35-46. Geneva: WHO. Available at: https://www.who.int/docstore/peh/noise/guidelines2.html

[3] National Sleep Foundation. (2023). *The Bedroom Environment: What Makes the Ideal Sleep Environment?* Sleep Foundation. Retrieved December 2024 from https://www.sleepfoundation.org/bedroom-environment

[4] Ancoli-Israel, S., Cole, R., Alessi, C., Chambers, M., Moorcroft, W., & Pollak, C. P. (2003). The role of actigraphy in the study of sleep and circadian rhythms. *Sleep*, 26(3), 342-392. https://doi.org/10.1093/sleep/26.3.342

[5] Buysse, D. J., Reynolds, C. F., Monk, T. H., Berman, S. R., & Kupfer, D. J. (1989). The Pittsburgh Sleep Quality Index: A new instrument for psychiatric practice and research. *Psychiatry Research*, 28(2), 193-213. https://doi.org/10.1016/0165-1781(89)90047-4

[6] Halperin, D. (2014). Environmental noise and sleep disturbances: A threat to health? *Sleep Science*, 7(4), 209-212. https://doi.org/10.1016/j.slsci.2014.11.003

[7] Pan, L., Feng, Q., & Stone, M. (2018). Effects of indoor humidity on human health: A systematic review. *Building and Environment*, 143, 106-116. https://doi.org/10.1016/j.buildenv.2018.06.055

Validation Strategy and Future Work

Phase 1: Rule-Based Classification (Current Implementation)

Deterministic scoring based on established research thresholds. Transparent, explainable algorithm for clinical review. Suitable for initial deployment and Basic-level project requirements. No training data required.

Advantages:

- Immediate deployment without data collection phase
- Consistent, reproducible classifications
- Directly traceable to scientific literature
- Clinically interpretable results

Phase 2: Machine Learning Enhancement (Future Work)

Proposed Methodology:

- **Data Collection:** Deploy system for 3-6 months, collect 100-300 nights of sensor data with patient-reported sleep quality (PSQI)

• **Model Training:** Random Forest Classifier with statistical aggregates, time-series features, and interaction terms

• **Performance Metrics:** Target accuracy >75%, sensitivity >70%, specificity >75%, correlation R^2 >0.6

• **Validation:** Compare ML vs. rule-based, clinical review, A/B testing

Phase 3: Continuous Improvement (Long-term)

• Active Learning: Incorporate new labeled data to refine model

• Personalization: Patient-specific models for individual sleep patterns

• Multi-night Analysis: Consider sleep quality trends over weeks/months

• Integration with Wearables: Combine environmental + physiological signals

Clinical Application Guidelines

Use Case: Screening tool for sleep environment optimization

Not a Replacement For: Polysomnography, clinical sleep studies, or medical diagnosis

Best Practice: Results should inform discussions with healthcare providers, not replace professional evaluation